MATH GUESS

Class 11 Punjab Boards

Note: We will do this Guess Paper in our Live Sessions

Examples are by far Important

Chapter # 01

1. Find the multiplicative inverse of each of the following complex numbers:

(i) $(-4, 7)$ (ii) $(\sqrt{2}, -\sqrt{5})$ (iii) $(1, 0)$

2. Separate into real and imaginary parts (write as a simple complex number):

(i) $\dfrac{2-7i}{4+5i}$ (ii) $\dfrac{(-2+3i)^2}{1+i}$ (iii) $\dfrac{i}{1+i}$ (iv) $\dfrac{(4+3i)^2}{4-3i}$

8. Find the least positive value of n, if $\left(\dfrac{1+i}{1-i}\right)^{2n} = 1$

1. **Find the real values of x and y in each of the following:**

(i) $x+iy+2-3i=i(5-i)(3+4i)$

(ii) $(x+iy)(1-i)=(2-3i)(-5+5i)(-i\frac{3}{5})$

(iii) $\dfrac{x}{2+i}+\dfrac{y}{3-i}=4+5i$

3. **Find the real values of x and y if:**

(i) $(x+iy)^2=25+60i$ (ii) $(x+iy)^2=64+48i$ (iii) $(x+iy)^2=\dfrac{2i-3}{3+i}$

7. **Find the square root of the following complex numbers:**

(i) $-7-24i$ (ii) $8-6i$ (iii) $-15-36i$ (iv) $119+120i$

9. **Find the real values of x and y if $(-7+i)(x+iy)+(-1-5i)=i(11-i)$**

10. **Find the real values of x and y if $(5-2i)(x+iy)+3=i(11-i)-4i$**

1. **Factorize the following:**

(i) a^2+4b^2 (ii) $9a^2+16b^2$ (iii) $3x^2+3y^2$ (iv) $144x^2+225y^2$

(v) $z^2-2iz-1$ (vi) $z^2+6z+13$ (vii) z^2+4z+5 (viii) $2z^2-22z+65$

2. **Factorize the following polynomials into its linear factors:**

(i) z^3+8 (ii) z^3+27 (iii) $z^3-2z^2+16z-32$ (iv) z^4+21z^2-100

(v) z^4-16 (vi) z^4+3z^2-4 (vii) z^4+5z^2+6 (viii) z^4-32z^2-3969

4. **Solve the following complex quadratic equations by completing square method:**

(i) $2z^2-3z+4=0$ (ii) $z^2-6z+30=0$ (iii) $3z^2-18z+50=0$

(iv) $z^2+4z+13=0$ (v) $2z^2+6z+9=0$ (vi) $3z^2-5z+7=0$

5. **Solve the following equations:**

(i) $2z^4-32=0$ (ii) $3z^5-243z=0$ (iii) $5z^5-5z=0$

(iv) $z^3-5z^2+z-5=0$ (v) $4z^4-25z^2-21=0$ (vi) $z^3+z^2+z+1=0$

1. Find the three cube roots of:

(i) 8 (ii) -8 (iii) -27 (iv) 64 (v) -125

3. If 1, ω, ω^2 are the cube roots of unity, show that $1+\omega^n+\omega^{2n}=3$ where n is a multiple of 3 respectively.

4. Evaluate:

(i) $\left(\frac{-1+\sqrt{-3}}{2}\right)^7+\left(\frac{-1-\sqrt{-3}}{2}\right)^7$ (ii) $(-1+\sqrt{-3})^5+(-1-\sqrt{-3})^5$

5. Show that $(1-\omega+\omega^2)(1-\omega^2+\omega^4)(1-\omega^4+\omega^8)(1-\omega^8+\omega^{16})\ldots$ to $2n$ factors $=2^{2n}$

6. Prove that $\left(\frac{i+\sqrt{3}}{2}\right)^8+\left(\frac{i-\sqrt{3}}{2}\right)^8=-1$.

9. If ω is a cube root of unity, prove that $\dfrac{a\omega^{12}+b\omega^{17}+c\omega^{19}}{a\omega^{14}+b\omega^{22}+c\omega^{30}}=\omega$

Chapter # 02

3. Express the following:

(a) The area A of a square as a function of its perimeter P.

(b) The circumference C of a circle as a function of its area A.

(c) The surface area S of a cube as a function of its volume V.

4. Find the domain and the range of the function g defined below:

(i) $g(x)=5-x$ (ii) $g(x)=\sqrt{x+2}$ (iii) $g(x)=\begin{cases}6x+7, & x\le -2\\ 4-3x, & x>-2\end{cases}$

(iv) $g(x)=|x-5|$ (v) $g(x)=\dfrac{x+2}{3-x}$

6. A stone falls from a height of 60m on the ground, the height h after x seconds is approximately given by $h(x) = 40 - 10x^2$.

   (i) What is the height of stone when:

   (a) $x = 1$ sec ? (b) $x = 1.5$ sec ? (c) $x = 1.7$ sec ?

   (ii) When does the stone strike the ground?

7. Consider the function $f(x) = 3x - 5$.

   (i) Determine the domain and range of $f(x)$.

   (ii) Is the function f one-to-one? Justify your answer.

   (iii) Is the function f onto if the co-domain is all real numbers? Explain.

1. Find the point of intersection of the coordinate axes and the following linear functions graphically:

   (i) $y = -5x + 10$ (ii) $y = 2x - 1$

   (iii) $y = \frac{1}{2}x - 3$ (iv) $y = 3x + \frac{3}{2}$

2. Find the point(s) of intersection of the following functions graphically:

   (i) $f(x) = 2x + 5$, $g(x) = -x + 5$

   (ii) $f(x) = 3x - 2$, $g(x) = 10 - x$

3. Graph the following functions:

   (i) $y = \sqrt{3x}$ (ii) $y = \sqrt{x} + 5$

   (iii) $y = -\frac{1}{2}\sqrt{x}$ (iv) $y = -\sqrt{x+1} + 2$

   (v) $y = \sqrt[3]{2x+1}$ (vi) $y = 2\sqrt[3]{x} - 3$

   (vii) $y = \sqrt[3]{x^2 + x - 2}$

Chapter # 03

1. Find the maximum or minimum value of the following quadratic functions by completing square:

(i) $f(x)=x^2+6x+13$ (ii) $f(x)=x^2+4x$

(iii) $f(x)=-x^2+8x+13$ (iv) $f(x)=-x^2-3x-5$

(v) $f(x)=3x^2+6x-13$ (vi) $f(x)=-2x^2-x+21$

3. Find the inverse of the following quadratic functions. Also find their domain and range:

(i) $f(x)=x^2-3,\ \ x\leq 0$ (ii) $f(x)=x^2+6x+4,\ \ x<-3$

(iii) $f(x)=2x^2-8x+11,\ \ x\geq 2$ (iv) $f(x)=3x^2-2x+6,\ \ x\geq 5$

(v) $f(x)=2(x-3)^2+1,\ \ x\geq 3$ (vi) $f(x)=-3(x+4)^2-5,\ \ x<-4$

4. Solve the following absolute value quadratic equations and inequalities:

(i) $|x^2+1|=5$ (ii) $|x^2+5x+4|=0$ (iii) $|x^2-6x+8|=4$

(iv) $|3x^2-7x+2|=x^2-x+1$ (v) $|x^2-4|<5$ (vi) $|x^2-3x+2|>4$

(vii) $|x^2-5x+6|\leq x+2$ (viii) $|2x^2-3x-5|<4$

EXERCISE 3.2

1. Solve the following equations:

(i) $\dfrac{1}{3x}+\dfrac{4x}{6}=1, x\neq 0$ (ii) $\dfrac{x}{x+1}+\dfrac{x+1}{x}=\dfrac{5}{2}; x\neq -1,0$

(iii) $\dfrac{1}{x+1}+\dfrac{2}{x+2}=\dfrac{7}{x+5}; x\neq -1,-2,-5$

(iv) $\dfrac{a}{ax-1}+\dfrac{b}{bx-1}=a+b; x\neq \dfrac{1}{a},\dfrac{1}{b}$ (v) $\dfrac{x+1}{x-1}+\dfrac{x-1}{x+1}=2, x\neq 1, x\neq -1$

(vi) $3x^2+15x-2\sqrt{x^2+5x+1}=2$ (vii) $\sqrt{2x+8}+\sqrt{x+5}=7$

(viii) $\sqrt{3x+4}=2+\sqrt{2x-4}$ (ix) $\sqrt{x+7}+\sqrt{x+2}=\sqrt{6x+13}$

(x) $\sqrt{x+5}-\sqrt{x-3}=2$

Chapter # 04

4. If A and B are square matrices of the same order, then explain why in general;
(i) $(A+B)^2 \neq A^2+2AB+B^2$ (ii) $(A-B)^2 \neq A^2-2AB+B^2$
(iii) $(A+B)(A-B) \neq A^2-B^2$

6. Solve the matrix equation $A^2-5A+4I-X=0$ if $A=\begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix}$

7. If A and B are two matrices such that $AB=B$ and $BA=A$, show that $A^2+B^2=A+B$.

2. Without expansion show that:

(i) $\begin{vmatrix} 7 & 8 & 9 \\ 5 & 6 & 7 \\ 2 & 3 & 4 \end{vmatrix}=0$ (ii) $\begin{vmatrix} 5 & 6 & -1 \\ 2 & 2 & 0 \\ 2 & -8 & 10 \end{vmatrix}=0$ (iii) $\begin{vmatrix} -a & 0 & b \\ 0 & a & -c \\ c & -b & 0 \end{vmatrix}=0$

(iv) $\begin{vmatrix} l & m+n & 1 \\ m & n+l & 1 \\ n & l+m & 1 \end{vmatrix}=0$ (v) $\begin{vmatrix} 2 & 1 & 3x \\ 2 & 3 & 9x \\ 3 & 5 & 15x \end{vmatrix}=0$ (vi) $\begin{vmatrix} bc & a & a^2 \\ ca & b & b^2 \\ ab & c & c^2 \end{vmatrix}=\begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix}$

3. Using properties of determinants, show that:

(i) $\begin{vmatrix} 3 & 5 & 0 \\ 5 & 25 & 10 \\ 7 & 25 & 1 \end{vmatrix}=25\begin{vmatrix} 3 & 1 & 0 \\ 1 & 1 & 2 \\ 7 & 5 & 1 \end{vmatrix}$ (ii) $\begin{vmatrix} a+x & a & a \\ a & a+x & a \\ a & a & a+x \end{vmatrix}=x^2(3a+x)$

(iii) $\begin{vmatrix} 1 & x & yz \\ 1 & y & zx \\ 1 & z & xy \end{vmatrix}=\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}$ (iv) $\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}=(x-y)(y-z)(z-x)$

(v) $\begin{vmatrix} 1 & 1 & 1 \\ a+1 & b+1 & c+1 \\ (a+1)^2 & (b+1)^2 & (c+1)^2 \end{vmatrix} = (a-b)(b-c)(c-a)$

(vi) $\begin{vmatrix} a^2+b^2 & c^2 & c^2 \\ a^2 & b^2+c^2 & a^2 \\ b^2 & b^2 & c^2+a^2 \end{vmatrix} = 4a^2b^2c^2$

(vii) $\begin{vmatrix} a & b & c \\ b+c & c+a & a+b \\ a+b & b+c & c+a \end{vmatrix} = a^3+b^3+c^3-3abc$

(viii) $\begin{vmatrix} a+t & a & a \\ b & b+t & b \\ c & c & c+t \end{vmatrix} = t^2(a+b+c+t)$

(ix) $\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} = (a+b+c)^3$

(x) $\begin{vmatrix} y+z & z+x & x+y \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix} = (x+y+z)(x-y)(y-z)(z-x)$

(xi) $\begin{vmatrix} 1 & 1 & 1 \\ a^2+1 & b^2+1 & c^2+1 \\ a^3+a & b^3+b & c^3+c \end{vmatrix} = (a-b)(b-c)(c-a)(ab+bc+ca-1)$

(xii) $\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = abc+ab+bc+ca$ (xiii) $\begin{vmatrix} 1 & a & a^2-bc \\ 1 & b & b^2-ca \\ 1 & c & c^2-ab \end{vmatrix} = 0$

8. Find the values of λ if A and B are singular.

$$A=\begin{bmatrix}4 & 2 & 3\\ 7 & \lambda & 6\\ 2 & 3 & 1\end{bmatrix}, \quad B=\begin{bmatrix}-2 & 4 & 5\\ 1 & -2 & 1\\ 2 & \lambda & 0\end{bmatrix}$$

9. Find the inverse of $A=\begin{bmatrix}1 & 2 & 1\\ -5 & 0 & 4\\ 5 & 4 & 0\end{bmatrix}$ and show that $A^{-1}A=I_3$

3. Solve the following systems of linear equations by Cramer's rule:

(i) $\left.\begin{aligned}2x+y-z&=1\\ x-y+2z&=3\\ 3x+2y+z&=4\end{aligned}\right\}$ (ii) $\left.\begin{aligned}x_1+2x_2-3x_3&=0\\ 4x_1-x_2+x_3&=5\\ -2x_1+3x_2+2x_3&=3\end{aligned}\right\}$ (iii) $\left.\begin{aligned}2x_1-x_2+x_3&=1\\ x_1+2x_2+2x_3&=2\\ x_1-2x_2-x_3&=1\end{aligned}\right\}$

4. Solve the following systems of linear equations by matrix inversion method:

(i) $\left.\begin{aligned}x-2y+z&=-1\\ 3x+y-2z&=4\\ y-z&=1\end{aligned}\right\}$ (ii) $\left.\begin{aligned}2x_1+x_2+3x_3&=3\\ x_1+3x_2-2x_3&=0\\ -3x_1-x_2+2x_3&=4\end{aligned}\right\}$ (iii) $\left.\begin{aligned}x+y&=2\\ 2x-z&=1\\ 2y-3z&=-1\end{aligned}\right\}$

Chapter # 05

EXERCISE 5.1

3. $\dfrac{x^2+1}{(x+1)(x-1)}$

10. $\dfrac{x+1}{(x-1)^2}$

11. $\dfrac{x^2+x}{(x^2-1)^2}$

EXERCISE 5.2

Resolve into partial fractions:

1. $\dfrac{2x^2+3x+3}{(x+1)(x^2+1)}$

2. $\dfrac{2x+1}{(x-2)(x^2+3x+5)}$

3. $\dfrac{2x+32}{(x-2)(x^2+2)^2}$

Chapter # 06

EXERCISE 6.2

6. Is 301 a term of the A.P. 5, 11, 17, ...?
7. If $2x, x+8, 3x+1$ are in A.P., then find the value of x.
8. Which term of the A.P., 3, 8, 13, ... is 123?
9. Which term of the A.P., 30, 29.5, 29, ... is the first negative term?
10. The 7^{th} and 21^{st} terms of an A.P., are 37 and 107 respectively. Find the A.P. and its 100^{th} term.
11. If $\frac{1}{a-c}, \frac{1}{b-c}, \frac{1}{b-a}$ are in A.P., the show that $\frac{a-b}{a-c}=\frac{a-c}{b-a}$.
17. If $\frac{1}{a}, \frac{1}{b}$ and $\frac{1}{c}$ are in A.P., show that $b=\frac{2ac}{a+c}$.
18. If $\frac{1}{a}, \frac{1}{b}$ and $\frac{1}{c}$ are in A.P., show that the common difference is $\frac{a-c}{2ac}$.

EXERCISE 6.3

2. If 6, 11, 16 are three A.Ms. between a and b, find a and b.
3. Insert five A.Ms. between $\sqrt{2}$ and $\frac{15}{\sqrt{2}}$.
4. The A.M. of two numbers is 7 and their product is 45. Find the numbers.
7. For what value of n, $\frac{a^{n+1}+b^{n+1}}{a^n+b^n}$ is the A.M. between a and b, where $a \neq b$.

EXERCISE 6.4

1. Sum the series:

(i) $3+6+9+\cdots+a_{20}$ (ii) $\frac{4}{\sqrt{5}}+\sqrt{5}+\frac{6}{\sqrt{5}}+\cdots+a_n$

4. How many terms of the series: $96+93+90+\cdots$ amount to 1071.
5. If the three sides of a right-angled triangle having perimeter 36 *cm* are in A.P., find them.
8. The 5^{th} and 9^{th} term of an A.P. are 11 and 17 respectively. Find the sum of 20 terms.
12. The sum of three numbers in an A.P. is 24 and their product is 440. Find the numbers.
13. The first four terms of an A.P. are 2, 6, 10 and 14. Find the least number of terms needed so that the sum of the terms is greater than 2000.
14. Find four numbers in A.P. whose sum is 32 and the sum of whose squares is 276.
15. Find the five numbers in A.P. whose sum is 25 and the sum of whose squares is 135.
16. If $\frac{1}{a+b}, \frac{1}{c+a}, \frac{1}{b+c}$ are in A.P. then show that a^2, b^2, c^2 are in A.P.

EXERCISE 6.5

1. Find the 6^{th} term of the *G.P.*: $-6, -3, \frac{-3}{2}, \cdots$.
2. Find the 8^{th} term of the sequence, $3, 3^2, 3^3, \cdots$.
6. If the 4^{th} and 9^{th} terms of a G.P. are 54 and 13122 respectively. Find the G.P. Also find its general term.
11. If $\frac{1}{a}, \frac{1}{b}$ and $\frac{1}{c}$ are in G.P. Show that the common ratio is $\pm\sqrt{\frac{a}{c}}$.
12. If the numbers 1, 4 and 3 are subtracted from three consecutive terms of an A.P., the resulting numbers are in G.P. Find the original numbers if their sum is 21.
13. If three consecutive numbers in A.P. are increased by 1, 4, 15 respectively, the resulting numbers are in G.P. Find the original numbers if their sum is 6.

EXERCISE 6.6

2. Insert four real geometric means between 3 and 96.
3. If both x and y are positive distinct real numbers, show that the geometric mean between x and y is less than their arithmetic mean.
4. For what value of n, $\frac{a^n+b^n}{a^{n-1}+b^{n-1}}$ is the positive geometric mean between a and b?
6. The A.M. between two numbers is 5 and their (positive) G.M. is 4. Find the numbers.

EXERCISE 6.7

1. Find the sum of first 15 terms of the G.P., 1, $\frac{1}{3}$, $\frac{1}{9}$, ...
2. The 3rd term of a G.P. is 16 and the 6th term is –128. Find the first term and the sum of the first seven terms.
3. Sum to n terms the series:

   (i) $0.2 + 0.22 + 0.222 + \cdots$ (ii) $3 + 33 + 333 + \cdots$

EXERCISE 6.9

1. Find the 9^{th} term of the following harmonic sequences:

(i) $\frac{1}{3}, \frac{1}{5}, \frac{1}{7}, \ldots$ (ii) $\frac{-1}{5}, \frac{-1}{3}, -1, \ldots$

2. Insert five harmonic means between the following given numbers:

(i) $\frac{-2}{5}$ and $\frac{2}{13}$ (ii) $\frac{1}{4}$ and $\frac{1}{24}$

3. The first term of an H.P. is $-\frac{1}{3}$ and the fifth term is $\frac{1}{5}$. Find its 9^{th} term.

4. If 5 is the harmonic mean between 2 and b, find b.

5. If the numbers $\frac{1}{k}, \frac{1}{2k+1}$ and $\frac{1}{4k-1}$ are in harmonic sequence, find k.

6. Find n so that $\frac{a^{n+1}+b^{n+1}}{a^n+b^n}$ may be H.M. between a and b.

8. If the H.M. and A.M. between two numbers are 4 and $\frac{9}{2}$ respectively, find the numbers.

9. If the (positive) G.M. and H.M. between two numbers are 4 and $\frac{16}{5}$, find the numbers.

Chapter # 07

EXERCISE 7.1

1. Evaluate each of the following:

(i) $\frac{10!}{0!\,8!}$ (ii) $\frac{12!}{3!(12-3)!}$ (iii) $\frac{1440}{3!\,4!} + \frac{2400}{5!\,2!}$ (iv) $\frac{(n+2)!}{(n+1)!}$

EXERCISE 7.2

2. Find the value of n when:

   (i) ${}^{n}P_3 = 504$ (ii) ${}^{15}P_n = 15 \cdot 14 \cdot 13 \cdot 12 \cdot 11$ (iii) ${}^{n}P_5 : {}^{n-2}P_2 = 540 : 1$

3. Prove from the first principle that:

   (i) ${}^{n}P_r = n \cdot {}^{n-1}P_{r-1}$ (ii) ${}^{n}P_r = {}^{n-1}P_r + r \cdot {}^{n-1}P_{r-1}$

4. How many words can be formed from the letters of the following words using all letters when no letter is to be repeated:

   (i) PYTHON (ii) NETWORK (iii) COMPUTER

5. How many signals can be given by 6 flags of different colours, using 2 flags at a time?

EXERCISE 7.3

1. How many arrangements of the letters of the following words, taken all together can be made?

   (i) PAKISTAN (ii) CURRICULUM (iii) PROBABILITY

2. How many permutations of the letters of the word "BANANA" can be made, if B must be the first letter in each arrangement?

3. How many arrangements of the letters of the word TRIGONOMETRY can be made, if each arrangement begins with T and ends with Y?

13. In how many ways can 8 keys be arranged in a circular key ring?

14. How many necklaces can be made from 10 beads of different colours?

EXERCISE 7.4

2. (i) If ${}^{3n}C_2 : {}^{n}C_2 = 15:1$, find n. (ii) If ${}^{n}P_r = 120$ and ${}^{n}C_r = 20$, find r.

3. Find the values of n and r, when

(i) ${}^{n}C_r = 56$, ${}^{n}P_r = 336$ (ii) ${}^{n-1}C_{r-1} : {}^{n}C_r : {}^{n}C_r = 1:3:7$

11. How many diagonals and triangles can be formed by joining the vertices of the polygon having 15 sides.

12. Find the number of sides of a polygon if the number of its diagonals is 104.

13. How many triangles can be formed by joining 15 points, 6 of which lie on the same straight line?

14. The members of a club are 10 boys and 8 girls. In how many ways can a committee of 6 boys and 3 girls be formed?

Chapter # 09

EXERCISE 9.1

5. Use synthetic division to find the quotient and the remainder when the polynomial $x^4 - 10x^2 - 2x + 4$ is divided by $x + 3$.

6. If $x + 1$ and $x - 2$ are factors of $x^3 - px^2 + qx + 2$. Using synthetic division, find the values of p and q.

7. When the polynomial $4x^4 + 2x^3 + kx^2 + 13$ is divided by $x + 1$, the remainder is 16. Find the value of k.

8. When the polynomial $x^3 + x^2 + x + k$ is divided by $x + 1$, the reminder is 7. Find the value of k.

EXERCISE 9.2

1. Consider a data set at monthly sales figures. A polynomial regression model $P(x)=x^3+2x^2+x-3$ is fitted to this data. If the observed sales in the 5^{th} month are 240 units, find the percentage error.

6. A signal processing system has a transfer function with a denominator $A(z) = z^2 - 0.3z - 0.4$. Use factor theorem to find the poles of the system and determine if the system is stable.

Chapter # 10

10.1.1 Fundamental Law of Trigonometry

Let α and β be any two angles (real numbers), then

$$\cos(\alpha-\beta)=\cos\alpha\ \cos\beta+\sin\alpha\ \sin\beta$$

which is called the **Fundamental Law of Trigonometry.**

EXERCISE 10.1

3. **Prove the following:**

(i) $\sin(180° + \alpha)\ \sin(90° - \alpha) = -\sin\alpha\ \cos\alpha$

(ii) $\sin 810°\ \sin 630° + \cos 135°\ \sin 225° = -\frac{1}{2}$

(iii) $\tan 150°\ \cot 330° - 2\sec 135°\ \text{cosec}\, 225° = -3$

(iv) $\sin 210° + \cos 240° + \tan 225° + \cot 225° = 1$

EXERCISE 10.2

2. **Prove that:** (i) $\sin(45° + \alpha) = \frac{1}{\sqrt{2}}(\sin\alpha + \cos\alpha)$

(ii) $\cos(\alpha + 45°) = \frac{1}{\sqrt{2}}(\cos\alpha - \sin\alpha)$

3. **Prove that:** (i) $\tan(45° + A)\tan(45° - A) = 1$

(ii) $\tan\left(\frac{\pi}{4} - \theta\right) + \tan\left(\frac{3\pi}{4} + \theta\right) = 0$ (iii) $\sin\left(\theta + \frac{\pi}{6}\right) + \cos\left(\theta + \frac{\pi}{3}\right) = \cos\theta$

(iv) $\dfrac{\sin\theta - \cos\theta\ \tan\frac{\theta}{2}}{\cos\theta + \sin\theta\ \tan\frac{\theta}{2}} = \tan\frac{\theta}{2}$ (v) $\dfrac{1 - \tan\theta\ \tan\phi}{1 + \tan\theta\ \tan\phi} = \dfrac{\cos(\theta + \phi)}{\cos(\theta - \phi)}$

8. If $\sin\alpha = \frac{24}{25}$ and $\cos\beta = \frac{20}{29}$, where $0 < \alpha < \frac{\pi}{2}$ and $0 < \beta < \frac{\pi}{2}$.

Show that $\sin(\alpha - \beta) = \frac{333}{725}$.

9. If $\sin\alpha = -\frac{8}{17}$ and $\cos\beta = -\frac{4}{5}$ where $\frac{3\pi}{2} < \alpha < 2\pi$ and $\pi < \beta < \frac{3\pi}{2}$. Find

(i) $\sin(\alpha+\beta)$ (ii) $\cos(\alpha+\beta)$ (iii) $\tan(\alpha+\beta)$

(iv) $\sin(\alpha-\beta)$ (v) $\cos(\alpha-\beta)$ (vi) $\tan(\alpha-\beta)$.

In which quadrants do the terminal sides of the angles of measures $(\alpha+\beta)$ and $(\alpha-\beta)$ lie?

10. Find $\sin(\alpha+\beta)$ and $\cos(\alpha+\beta)$, given that

(i) $\tan\alpha = \frac{3}{4}$, $\cos\beta = \frac{5}{13}$ and neither the terminal side of the angle of measure α nor that of β is in the quadrant I.

(ii) $\tan\alpha = -\frac{15}{8}$ and $\sin\beta = -\frac{7}{25}$ and neither the terminal side of the angle of measure α nor that of β is in the quadrant IV.

11. Prove that: $\frac{\cos 19^\circ + \sin 19^\circ}{\cos 19^\circ - \sin 19^\circ} = \tan 64^\circ$.

EXERCISE 10.4

1. Express the following products as sums or differences:

(i) $2\sin 3\theta \cos\theta$ (ii) $2\cos 5\theta \sin 3\theta$

(iii) $\sin 5\theta \cos 2\theta$ (iv) $2\sin 7\theta \sin 2\theta$

(v) $\cos(x+y)\sin(x-y)$ (vi) $\cos(2x+30^\circ)\cos(2x-30^\circ)$

(vii) $\sin 12^\circ \sin 46^\circ$ (viii) $\sin(x+45^\circ)\sin(x-45^\circ)$

2. Express the following sums or differences as products:

(i) $\sin 5\theta + \sin 3\theta$ (ii) $\sin 8\theta - \sin 4\theta$

(iii) $\cos 6\theta + \cos 3\theta$ (iv) $\cos 7\theta - \cos\theta$

(v) $\cos 12^\circ + \cos 48^\circ$ (vi) $\sin(x+30^\circ) + \sin(x-30^\circ)$

5. Prove that:

(i) $\cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ = \frac{1}{16}$

(ii) $\sin\frac{\pi}{9}\sin\frac{2\pi}{9}\sin\frac{\pi}{3}\sin\frac{4\pi}{9} = \frac{3}{16}$

(iii) $\sin 10^\circ \sin 30^\circ \sin 50^\circ \sin 70^\circ = \frac{1}{16}$

Chapter # 11

EXERCISE 11.1

1. Determine whether the following functions are even, odd or neither odd nor even.

(i) $\sin^2 x$ (ii) $\sin x+\cos x$ (iii) $\sin^4 x+\cos^4 x$

(iv) $\tan x+\sec x$ (v) $\dfrac{1}{\operatorname{cosec}^3 x}$ (vi) $\dfrac{\sin x+\sin 3x}{\cos x+\cos 3x}$

(vii) $\dfrac{1}{\sec x+\sec^3 x}$ (viii) $\dfrac{1}{\sec x+\cot^2 x}$

2. Find the periods of the following functions:

(i) $\sin 5x$ (ii) $\cos 7x$ (iii) $\tan 3x$ (iv) $\cot \dfrac{x}{2}$

(v) $19\sin\left(\dfrac{\pi}{20}x\right)$ (vi) $\operatorname{cosec}\left(\dfrac{2x}{5}\right)$ (vii) $\dfrac{1}{2}\sin\left(\dfrac{3x}{2}-\dfrac{\pi}{2}\right)$

(viii) $-5-3\sec\left(7\pi x+\dfrac{\pi}{4}\right)$ (ix) $12+10\tan\left(\dfrac{\pi}{30}x\right)$

(x) $6-4\cot\left(\dfrac{7x}{4}+\dfrac{\pi}{4}\right)$ (xi) $9+30\sec\left(\dfrac{x}{15}+\dfrac{2\pi}{15}\right)$

EXERCISE 11.3

1. Find the maximum and minimum values of the following functions:

(i) $3-\sin 3x$ (ii) $3+\sin 2x$ (iii) $\dfrac{1}{2}+\sin(5x+\pi)$

(iv) $\dfrac{3}{2}+\cos\left(x-\dfrac{\pi}{4}\right)$ (v) $1-3\cos 2x$ (vi) $1+2\sin\left(x+\dfrac{\pi}{6}\right)$

(vii) $\dfrac{1}{10-2\sin 3x}$ (viii) $\dfrac{1}{7+3\cos(-2x)}$ (ix) $\dfrac{1}{5-3\cos(3x-1)}$

Chapter # 12

EXERCISE 12.1

3. **Evaluate each limit by using algebraic techniques:**

(i) $\lim\limits_{x\to -1}\dfrac{x^3-x}{x+1}$ (ii) $\lim\limits_{x\to 3}\left(\dfrac{x^2-5x+6}{x^2-2x-3}\right)$ (iii) $\lim\limits_{x\to 2}\left(\dfrac{x^3-8}{x^2-5x+6}\right)$

(iv) $\lim\limits_{x\to 1}\dfrac{x^3-3x^2+3x-1}{x^3-x}$ (v) $\lim\limits_{x\to 2}\left(\dfrac{x^3-6x^2+12x-8}{x^3-4x}\right)$ (vi) $\lim\limits_{x\to 1}\left(\dfrac{x^4-1}{x^2-3x+2}\right)$

(vii) $\lim\limits_{x\to 2}\dfrac{x-2}{\sqrt{x+2}-\sqrt{6-x}}$ (viii) $\lim\limits_{h\to 0}\dfrac{\sqrt{x+h}-\sqrt{x}}{h}$ (ix) $\lim\limits_{x\to a}\dfrac{x^n-a^n}{x^m-a^m}$

4. **Evaluate the following limits:**

(i) $\lim\limits_{x\to 0}\dfrac{\sin 5x}{x}$ (ii) $\lim\limits_{x\to 0}\dfrac{\sin x^{\circ}}{x}$ (iii) $\lim\limits_{\theta\to 0}\dfrac{1-\cos\theta}{\sin\theta}$

(iv) $\lim\limits_{x\to\frac{\pi}{4}}\dfrac{\sin x-\cos x}{x-\frac{\pi}{4}}$ (v) $\lim\limits_{x\to 0}\dfrac{\cos ax-\cos bx}{x^2}$ (vi) $\lim\limits_{x\to\frac{\pi}{4}}\dfrac{\tan x-1}{x-\frac{\pi}{4}}$

(vii) $\lim\limits_{x\to 0}\dfrac{1-\cos 2x}{x^2}$ (viii) $\lim\limits_{x\to 0}\dfrac{\cos ax-\cos bx}{\cos cx-\cos dx}$ (ix) $\lim\limits_{x\to 1}\dfrac{x^3-1}{x^2-1}$

(x) $\lim\limits_{x\to 3}\dfrac{x^2-x\log x+3\log x-9}{x-3}$ (xi) $\lim\limits_{x\to 0}\dfrac{x(2^x-1)}{1-\cos x}$

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5. **Express each limit in terms of e.**

(i) $\lim\limits_{n\to+\infty}\left(1+\dfrac{1}{n}\right)^{2n}$ (ii) $\lim\limits_{n\to+\infty}\left(1+\dfrac{1}{n}\right)^{\frac{n}{2}}$ (iii) $\lim\limits_{n\to+\infty}\left(1-\dfrac{1}{n}\right)^{n}$

(iv) $\lim\limits_{n\to+\infty}\left(1+\dfrac{1}{3n}\right)^{n}$ (v) $\lim\limits_{n\to+\infty}\left(1+\dfrac{4}{n}\right)^{n}$ (vi) $\lim\limits_{x\to 0}(1+3x)^{\frac{2}{x}}$

(vii) $\lim\limits_{x\to 0}(1+2x^2)^{\frac{1}{x^2}}$ (viii) $\lim\limits_{x\to 0}\dfrac{e^{ax}-e^{bx}}{abx}$ (ix) $\lim\limits_{x\to\infty}\left(\dfrac{x}{1+x}\right)^{x}$

(x) $\lim\limits_{x\to 0}\dfrac{e^{\frac{1}{x}}-1}{e^{\frac{1}{x}}+1},\ x<0$ (xi) $\lim\limits_{x\to 0}\dfrac{e^{\frac{1}{x}}-1}{e^{\frac{1}{x}}+1},\ x>0$ (xii) $\lim\limits_{x\to 2}\dfrac{e^x-e^2}{x-2}$

EXERCISE 12.2

2. Discuss the continuity of $f(x)$ at $x = c$

(i) $f(x)=\begin{cases} 2x+5 & \text{if } x\le 2 \\ 4x+1 & \text{if } x>2 \end{cases}$, $c=2$ (ii) $f(x)=\begin{cases} 3x-1 & \text{if } x<1 \\ 4 & \text{if } x=1 \\ 2x & \text{if } x>1 \end{cases}$, $c=1$

3. If $f(x)=\begin{cases} 3x & \text{if} & x\le -2 \\ x^2-1 & \text{if} & -2<x<2 \\ 3 & \text{if} & x\ge 2 \end{cases}$ Discuss continuity at $x=2$ and $x=-2$.

5. Find the values of m and n, so that given function f is continuous at $x=3$

(i) $f(x)=\begin{cases} mx & \text{if} & x<3 \\ n & \text{if} & x=3 \\ -2x+9 & \text{if} & x>3 \end{cases}$ (ii) $f(x)=\begin{cases} mx & \text{if} & x<3 \\ x^2 & \text{if} & x\ge 3 \end{cases}$

6. $f(x)=\begin{cases} \dfrac{\sqrt{2x+5}-\sqrt{x+7}}{x-2} & , & x\ne 2 \\ k & , & x=2 \end{cases}$

Find value of k so that f is continuous $x=2$.

Chapter # 13

EXERCISE 13.1

1. Find by definition, the derivatives w.r.t. 'x' of the following functions defined as:

   (i) $2x^2+1$ (ii) $2-\sqrt{x}$ (iii) $\dfrac{1}{\sqrt{x}}$ (iv) $x(x-3)$

2. Find $\dfrac{dy}{dx}$ from first principle and find gradient of the curve at the given point:

   (i) $\sqrt{x+2}$ at $x=6$ (ii) $\dfrac{1}{\sqrt{x+a}}$ at $x=a$

5. Find the gradient and equation of the tangent line to $y=3x^2-4x+1$ at $x=2$.
6. For the function $f(x)=2x^3+x$, calculate the equation of the tangent line at $x=-1$.
7. Find the coordinates of the point of tangency and the equation of the tangent line for $f(x)=x^3-2x+1$ at $x=1$.
8. Find the gradient of the curve $f(x)=3x^2+2x$ at $x=1$.

EXERCISE 13.2

1. Differentiate w.r.t 'x'.

(i) $x^4+2x^3+x^2$ (ii) $x^{-3}+2x^{-\frac{3}{2}}+3$ (iii) $\frac{2x-3}{2x+1}$

(iv) $\frac{(1+\sqrt{x})(x-x\ )}{\sqrt{x}}$ (v) $\left(\sqrt{x}-\frac{1}{\sqrt{x}}\right)^2$ (vi) $(x-5)(3-x)$

(vii) $\frac{(x^2+1)^2}{x^2-1}$ (viii) $\frac{x^2+1}{x^2-3}$ (ix) $\frac{2x-1}{\sqrt{x^2+1}}$

(x) $\sqrt{\frac{a-x}{a+x}}$ (xi) $\frac{\sqrt{x^2+1}}{\sqrt{x^2-1}}$

2. Find $\frac{dy}{dx}$ if $y=\frac{(\sqrt{x}+1)(x^{\frac{3}{2}}-1)}{x^{\frac{1}{2}}-1}$, $(x\neq 1)$

3. Differentiate $\frac{(\sqrt{x}+1)(x^{\frac{3}{2}}-1)}{x^{\frac{3}{2}}-x^{\frac{1}{2}}}$ with respect to x.

4. If $y=\sqrt{x}-\frac{1}{\sqrt{x}}$, show that $2x\frac{dy}{dx}+y=2\sqrt{x}$

5. If $y=x^4+2x^2+2$, prove that $\frac{dy}{dx}=4x\sqrt{y-1}$.

Chapter # 14

EXERCISE 14.1

3. Find t, so that $|2\underline{i}+(t-1)\underline{j}+t\underline{k}|=\sqrt{13}$

4. Find a unit vector in the direction of $\underline{v}=-\underline{i}+4\underline{j}-8\underline{k}$

5. If $\underline{u}=2\underline{i}+\underline{j}-3\underline{k}$, $\underline{v}=-\underline{i}+4\underline{j}+2\underline{k}$ and $\underline{w}=3\underline{i}-2\underline{j}+\underline{k}$, Find a unit vector parallel to $4\underline{u}-3\underline{v}+2\underline{w}$.

6. Find a vector whose
   (i) magnitude is 5 and is parallel to $3\underline{i}+4\underline{j}-\underline{k}$
   (ii) magnitude is 7 and is parallel to $-\underline{i}+\underline{j}+\underline{k}$.

11. Find the direction cosines for the given vector:
    (i) $\underline{u}=-6\underline{i}+3\underline{j}+2\underline{k}$ (ii) $\underline{v}=4\underline{i}+2\underline{j}-5\underline{k}$

EXERCISE 14.2

3. If $|\underline{a}|=3$, $|\underline{b}|=4$ and $|\underline{a}+\underline{b}|=5$. Find the angle between $\underline{a}$ and $\underline{b}$.

4. Calculate the projection of $\underline{a}$ along $\underline{b}$ and projection of $\underline{b}$ along $\underline{a}$ when:
   (i) $\underline{a}=2\underline{i}+3\underline{j}-\underline{k}$, $\underline{b}=\underline{i}-2\underline{j}+4\underline{k}$ (ii) $\underline{a}=4\underline{i}-2\underline{j}+3\underline{k}$, $\underline{b}=\underline{i}+\underline{j}+\underline{k}$

5. Find a real number α so that the vectors $\underline{u}$ and $\underline{v}$ are perpendicular:
   (i) $\underline{u}=\alpha\underline{i}+3\underline{j}+\underline{k}$, $\underline{v}=\underline{i}-2\underline{j}+\alpha\underline{k}$ (ii) $\underline{u}=\alpha\underline{i}+2\alpha\underline{j}-\underline{k}$, $\underline{v}=\underline{i}+\alpha\underline{j}+3\underline{k}$

10. Prove that the $\cos(\alpha+\beta)=\cos\alpha\cos\beta-\sin\alpha\sin\beta$

11. Prove that in any triangle ABC.
    (i) $b=c\cos A+a\cos C$ (ii) $c=a\cos B+b\cos A$
    (iii) $b^2=c^2+a^2-2ca\cos B$ (iv) $c^2=a^2+b^2-2ab\cos C$

13. Find the work done, if the point at which the constant force $\underline{F}=2\underline{i}+5\underline{j}+3\underline{k}$ is applied to an object, moves it from $P_1(2,-3,1)$ to $P_2(7,5,3)$.

14. A particle, acted by constant forces $\underline{F_1}=3\underline{i}+4\underline{j}-3\underline{k}$ and $\underline{F_2}=\underline{i}+4\underline{j}-\underline{k}$, is displaced from $A(2,1,3)$ to $B(5,4,4)$. Find the work done.

EXERCISE 14.3

3. Find the area of the triangle, formed by the points P, Q and R.

 (i) $P(2, 3, 5)\,;\,Q(1, 2, 0)\,;\,R(4, 1, 2)$ (ii) $P(0, 0, 1)\,;\,Q(2, -1, 2)\,;\,R(-1, 3, 2)$

4. Find the area of a parallelogram, whose vertices are:

 (i) $A(1, 1, 1)\,;\,B(4, 2, 3)\,;\,C(5, 6, 7)\,;\,D(2, 5, 5)$

 (ii) $A(4, 5, 6)\,;\,B(1, 3, 2)\,;\,C(-2, 0, 1)\,;\,D(1, 2, 5)$

5. If the cross product of the vectors $\underline{u} = 7\underline{i} - 4\underline{j} + 5\underline{k}$ and $\underline{v} = a\underline{i} - b\underline{j} + 3\underline{k}$ is zero, then find the values of a and b.

6. Which vectors, if any, are perpendicular or parallel

 (i) $\underline{u} = 5\underline{i} - \underline{j} + \underline{k}\,;\,\underline{v} = \underline{j} - 5\underline{k}\,;\,\underline{w} = -15\underline{i} + 3\underline{j} - 3\underline{k}$

 (ii) $\underline{u} = \underline{i} + 2\underline{j} - \underline{k}\,;\,\underline{v} = -\underline{i} + \underline{j} + \underline{k}\,;\,\underline{w} = -\dfrac{\pi}{2}\underline{i} - \pi\underline{j} + \dfrac{\pi}{2}\underline{k}$

8. Prove that: $\underline{a}\times(\underline{b}+\underline{c}) + \underline{b}\times(\underline{c}+\underline{a}) + \underline{c}\times(\underline{a}+\underline{b}) = \underline{0}$.

9. If $\underline{a}+\underline{b}+\underline{c}=\underline{0}$, then prove that $\underline{a}\times\underline{b} = \underline{b}\times\underline{c} = \underline{c}\times\underline{a}$

10. Prove that: $\sin(\alpha-\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$

11. Show that $|\underline{a}\times\underline{b}|^2 = |\underline{a}|^2|\underline{b}|^2 - (\underline{a}\cdot\underline{b})^2$

13. Find the moment about the point $M(1, -3, 3)$ of the force represented by $\overrightarrow{AB}$, where the coordinates of points $A(4, 3, -1)$ and $B(-1, 3, 7)$ are given.

14. A force $\overrightarrow{F} = 6\underline{i} + 4\underline{j} - 4\underline{k}$ is applied at the point $A(1,-1,2)$. Find the moment of the force about the point $B(3,-2,3)$.

15. Give a force $\underline{F} = 2\underline{i} + \underline{j} - 3\underline{k}$ acting at a point $A(1,-2,1)$. Find the moment of $\overrightarrow{F}$ about the point $B(2, 0, -2)$.

16. A force $\underline{F} = -2\underline{i} + \underline{j} - 3\underline{k}$ is applied at $P(-1, -3, 2)$. Find its moment about the point $Q(4, 2, 2)$.